IBM Data Science Capstone Project

SpaceX Falcon 9 Landing Prediction

End-to-end data science workflow: API data collection, web scraping, wrangling, SQL analysis, EDA, Folium maps, Plotly Dash dashboard and classification modeling.

Prepared by Omkar Sawant

https://github.com/OmkarSawant23/Data-Science-Capstone

4	90	4	83.3%
Launch sites	Falcon 9 records	ML models	Best test accuracy

Presentation Roadmap

Executive Summary

Objective: predict whether the Falcon 9 first stage will land successfully.
Collected and prepared launch data using SpaceX API, web scraping, data wrangling and SQL.
Explored patterns with Seaborn/Matplotlib, Folium maps and a Plotly Dash dashboard.
Built classification models using standardized features and GridSearchCV.
Best test accuracy reached 83.3%, with Logistic Regression, SVM and KNN tying on the test set.

Business value

Landing prediction helps estimate launch cost because booster recovery is directly linked to reusability and mission economics.

Analytical value

Combining SQL, EDA, geospatial analysis and ML provided stronger evidence than relying on one method alone.

Key insight

Success generally improved over time, and launch outcome is influenced by launch site, orbit type, payload mass and booster category.

Repository

All notebooks, datasets, Dash app and outputs are stored on GitHub.

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Introduction: Problem Context

SpaceX Falcon 9 first-stage reuse is a major factor behind lower launch costs.
A successful landing reduces the cost of future launches by enabling booster reuse.
The core data science question: can historical launch information predict first-stage landing success?
The project follows a full workflow from data acquisition to machine learning and business interpretation.

Collect	Analyze	Visualize	Predict
API + web scraping	SQL + EDA	Folium + Dash	Classification models

Target variable

Class = 1 for successful first-stage landing; Class = 0 for failed or non-successful landing outcomes.

GitHub Repository and Deliverables

notebooks

Completed Jupyter notebooks for API, scraping, wrangling, SQL, EDA, maps and ML

datasets

CSV files and SQLite database used across the workflow

dash_app

Plotly Dash application file

outputs

Folium HTML maps, dashboard screenshots and visual evidence

presentation

Final PowerPoint and PDF report

The repository is structured to support reproducibility and review: code, data, dashboard assets and visual outputs are organized in separate folders.

Data Collection and Data Wrangling Methodology

1. API collection

SpaceX REST API used to collect launch records, rocket IDs, payload data, launchpad metadata and landing outcomes.

2. Web scraping

BeautifulSoup used to extract Falcon 9 launch records from historical web tables.

3. Cleaning

Falcon 1 records removed, Falcon 9 records retained, missing PayloadMass values filled using the mean.

4. Label creation

Landing outcomes transformed into binary Class label for predictive modeling.

Output datasets: dataset_part_1.csv, spacex_web_scraped.csv, dataset_part_2.csv

EDA and Feature Engineering Methodology

Visual exploration used Matplotlib and Seaborn scatter plots, bar charts and trend lines.
Core relationships tested: launch site, orbit type, payload mass, flight number and landing success.
Yearly launch success trend was created to check operational learning over time.
Categorical variables were converted using one-hot encoding to support ML modeling.

Raw launch data → **EDA variables** → **One-hot encoding** → **ML feature set**

Final EDA output

dataset_part_3.csv prepared for classification models.

SQL Analysis Methodology

SQLite database my_data1.db was used to query the SpaceX launch dataset.
SQL queries applied filtering, grouping, aggregation, date extraction and sorting.
Analysis focused on launch site distribution, payload mass, booster versions, mission outcomes and landing outcome counts.

SELECT DISTINCT

Unique launch sites

GROUP BY

Mission outcome counts

AVG / SUM

Payload mass summaries

WHERE + DATE

Time-specific landing outcomes

Why SQL matters

SQL provided reproducible, question-driven validation of the same dataset before visualization and modeling.

Interactive Visual Analytics Methodology

Folium geospatial analysis

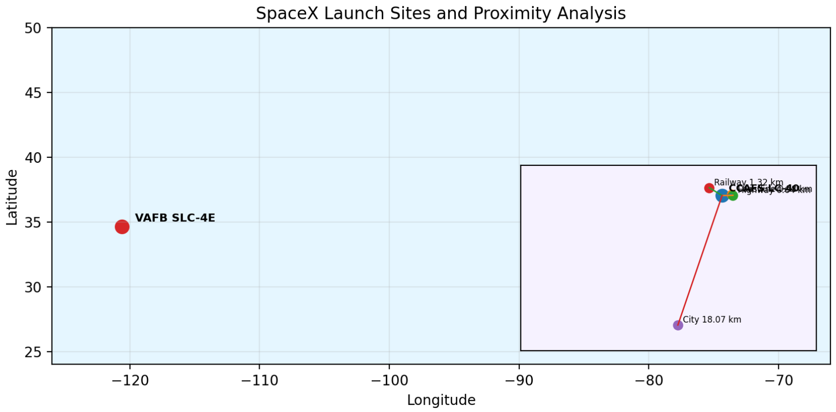
Interactive launch-site maps were created to show launch locations, clustered success/failure markers and proximity to coastlines, cities, railway and highway features.

Plotly Dash dashboard

A Dash web application was built with a launch site dropdown, payload range slider, pie chart and payload-success scatter plot.

User interaction value

Users can filter sites and payload ranges to visually compare landing success patterns in real time.



Predictive Analysis Methodology

Feature matrix

One-hot encoded launch site, orbit, landing pad and booster serial features

Target

Class label representing first-stage landing success

Preprocessing

StandardScaler applied to numerical feature matrix

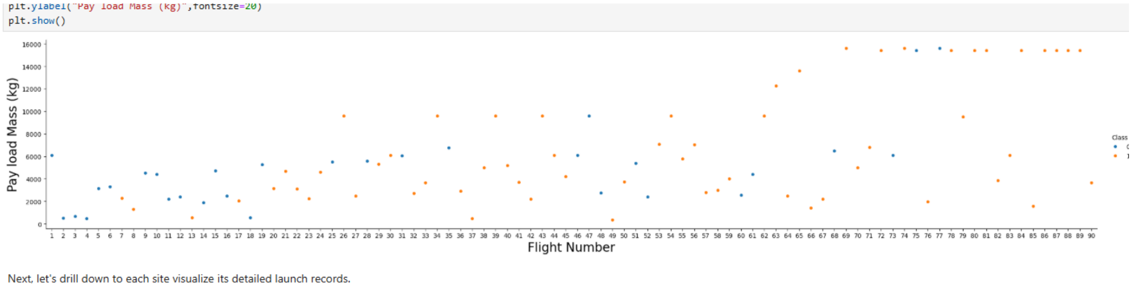
Evaluation

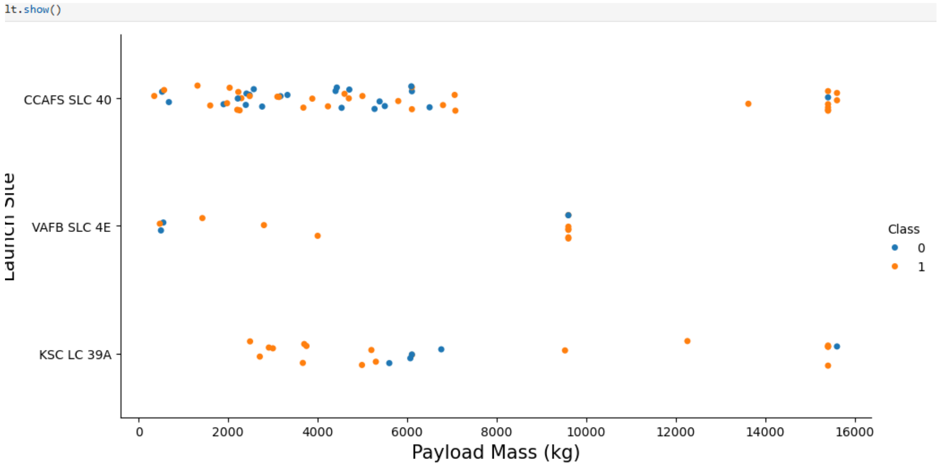
Train-test split, GridSearchCV and test accuracy comparison

Models compared

- **Logistic Regression**
- **Support Vector Machine**
- **Decision Tree**
- **K-Nearest Neighbors**

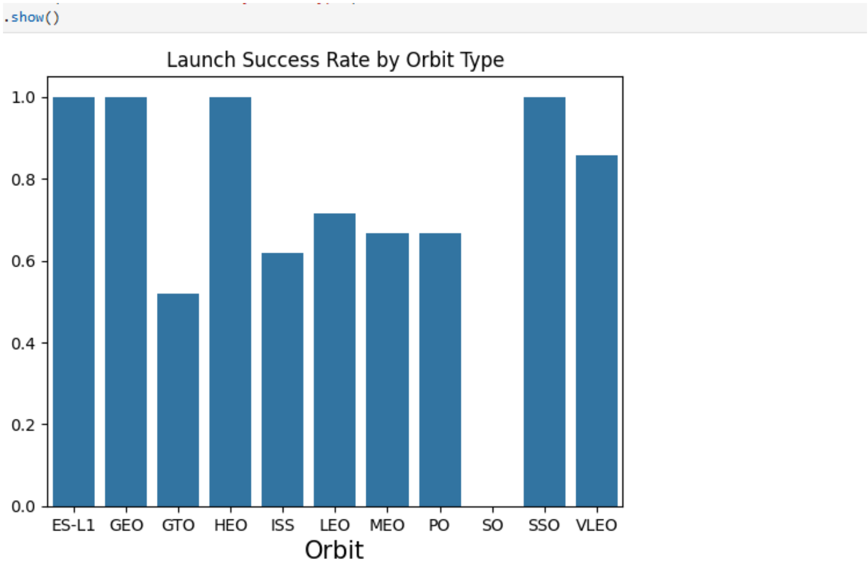
EDA Visualization Results: Launch Success Drivers

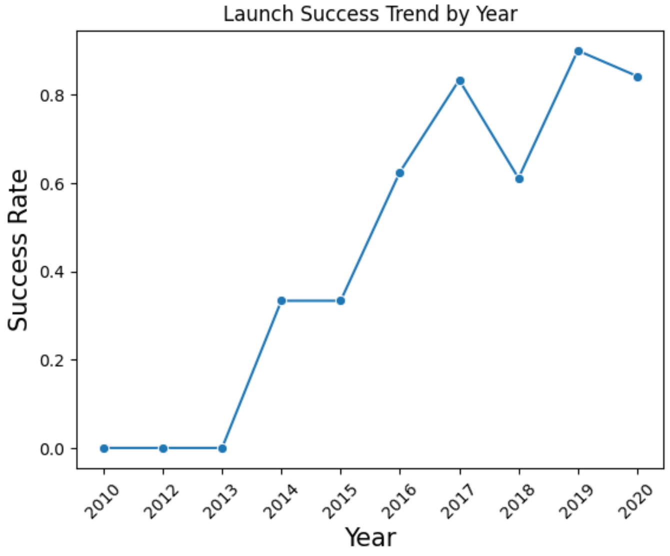




Flight number shows a visible relationship with success: later launches contain more successful landings, suggesting operational learning.
Payload mass alone does not fully determine success; successful landings occur across both lower and higher payload ranges.
Launch site patterns differ, showing that site-specific operations and mission profiles may affect landing outcomes.

EDA Visualization Results: Orbit and Time Trends





Orbit result

Some orbit categories show very high success rates, while GTO and ISS missions show more mixed outcomes.

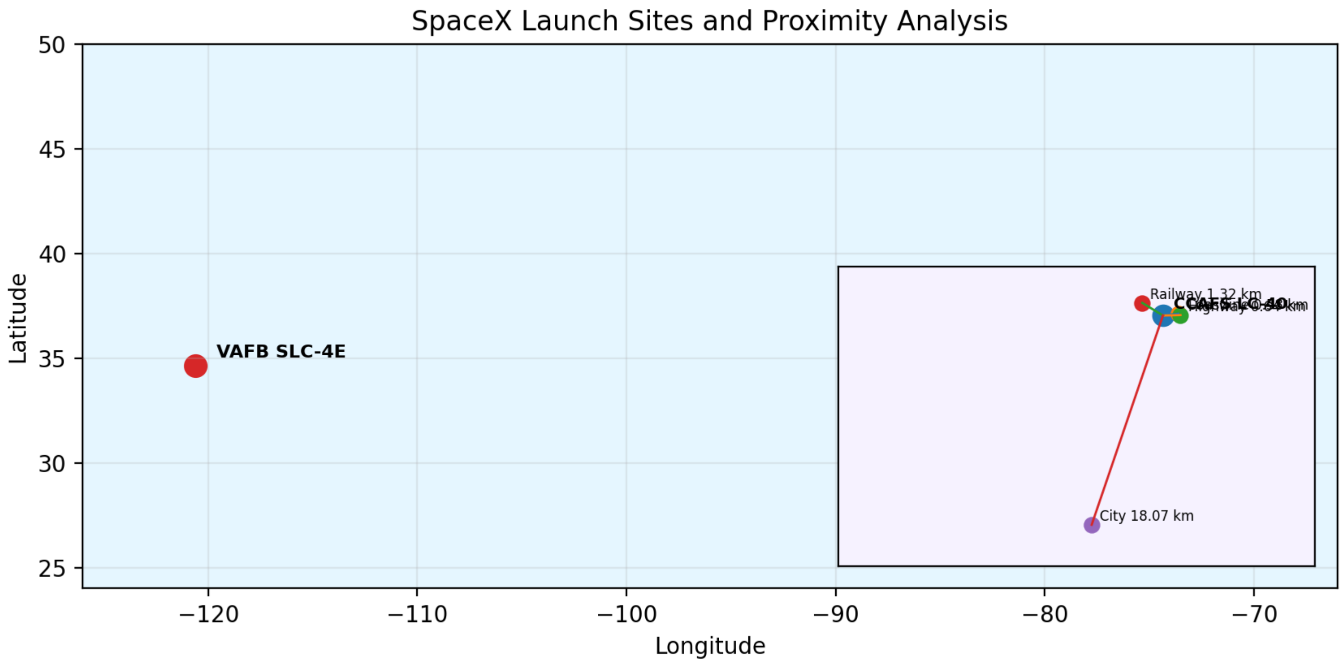
Time trend result

Success rate rises sharply after 2015 and remains comparatively strong in later years, supporting the learning-curve interpretation.

EDA with SQL Results

SQL question	Result / insight
Distinct launch sites	Four major sites were identified: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E.
Payload analysis	SQL aggregation helped summarize payload mass by customer, booster and mission type.
Landing outcomes	Mission and landing outcomes were grouped to compare success, failure, no-attempt and drone ship/ground pad outcomes.
First ground-pad success	The first successful ground pad landing occurred in December 2015.
Maximum payload missions	Several missions reached the maximum observed payload level of about 9,600 kg.

Interactive Map Results Using Folium

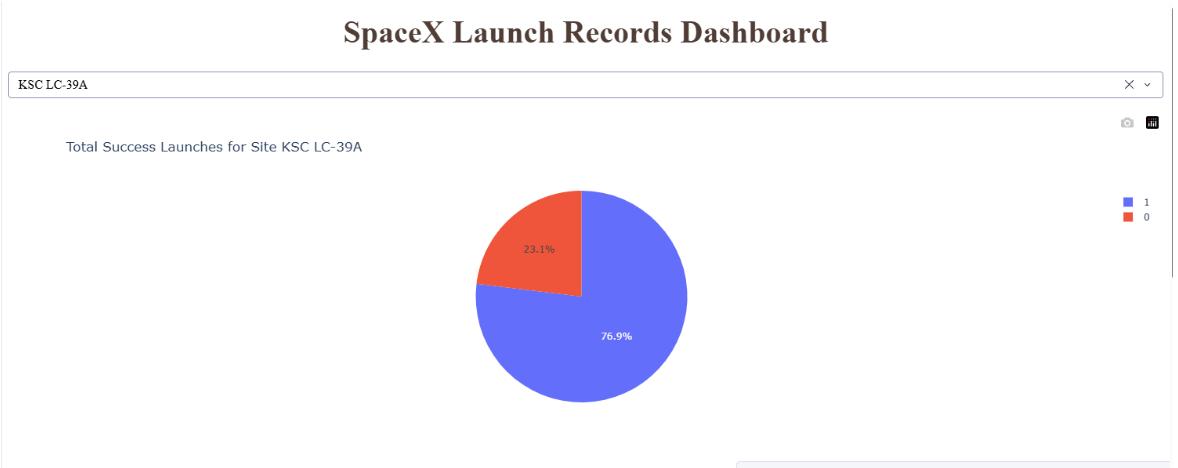


Launch sites are positioned near coastlines, which supports safety and stage recovery logistics.
Marker clusters distinguish successful and failed landings using green/red markers.
Distance analysis measured proximity from launch site to coastline, city, railway and highway points.
Example CCAFS LC-40 proximities: coastline ~0.58 km, railway ~1.32 km, highway ~0.64 km, city ~18.07 km.

Deliverable

Interactive HTML maps saved in outputs/.

Plotly Dash Dashboard Results



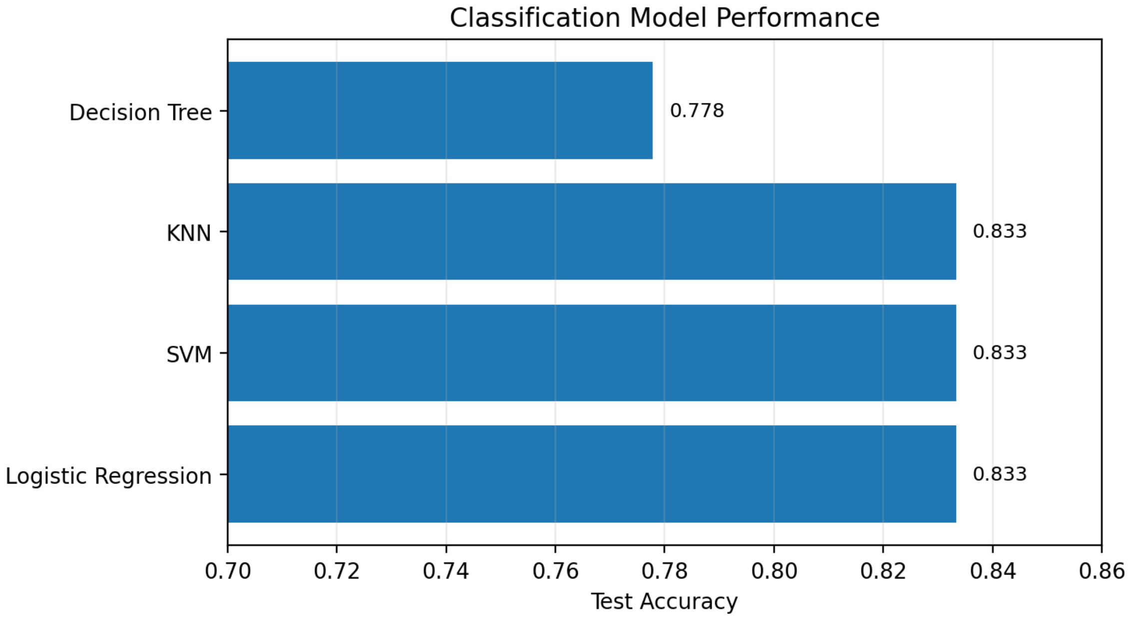
Launch site filter

Selecting KSC LC-39A showed a strong success share of 76.9% in the pie chart.

Payload slider

The scatter plot allows interactive filtering of payload mass and comparison by booster version category.

Predictive Analysis Results: Classification Models



	Model	Test Accuracy
0	Logistic Regression	0.833333
1	Support Vector Machine	0.833333
3	K Nearest Neighbors	0.833333
2	Decision Tree	0.777778

Logistic Regression, Support Vector Machine and KNN tied for the highest test accuracy at 83.3%.
Decision Tree produced a lower test accuracy of 77.8%, suggesting slightly weaker generalization on the test split.
Because three models tied, a simpler model such as Logistic Regression is a practical final choice for interpretability and deployment.

Innovative Insights and Business Implications

Learning curve

Success rates improved in later years, suggesting operational refinement and booster recovery learning.

Mission context matters

Launch site, orbit type and payload mass interact; no single variable fully explains success.

Booster category signal

Dash scatter plot shows booster categories are useful for visual comparison of landing outcomes.

Geospatial advantage

Coastal siting supports safe launch paths and recovery logistics while staying connected to nearby infrastructure.

Decision support

Combining dashboard, maps and ML creates a richer tool for estimating mission risk and launch cost.

Model choice

A tied top accuracy favors the simpler, more interpretable Logistic Regression model as a baseline.

Conclusion

The full SpaceX Falcon 9 workflow was completed from data collection through final predictive modeling.
EDA, SQL and visual analytics showed clear relationships between launch success and time, orbit, launch site, payload mass and booster category.
Folium and Dash added interactive exploration beyond static charts.
Classification models predicted first-stage landing success with a best test accuracy of 83.3%.
The project demonstrates how data science can support launch cost estimation and mission planning.

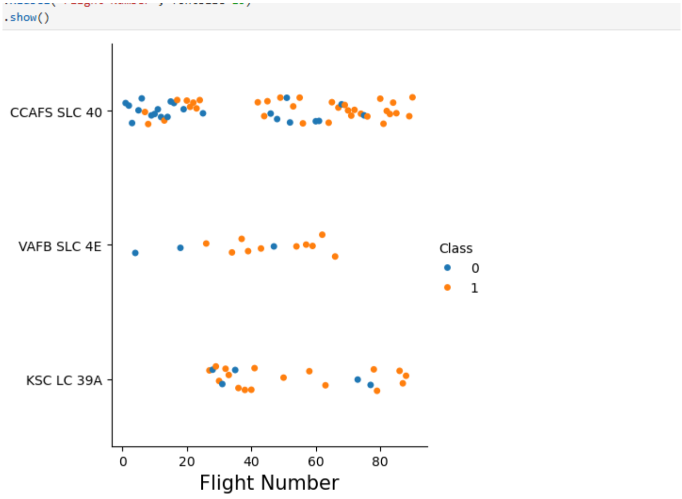
Final recommendation

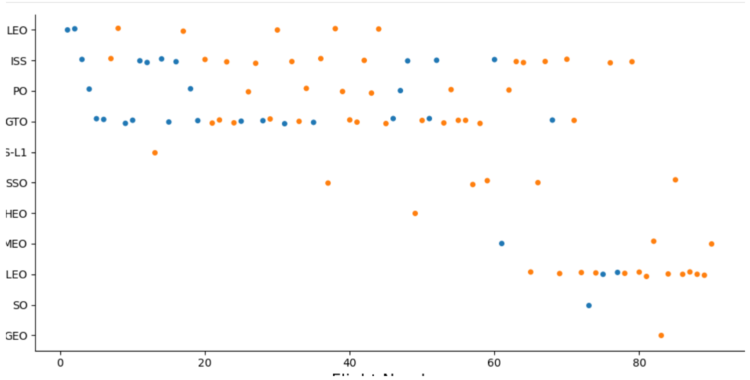
Use Logistic Regression as an interpretable baseline, while continuing to test larger datasets and more recent launch records.

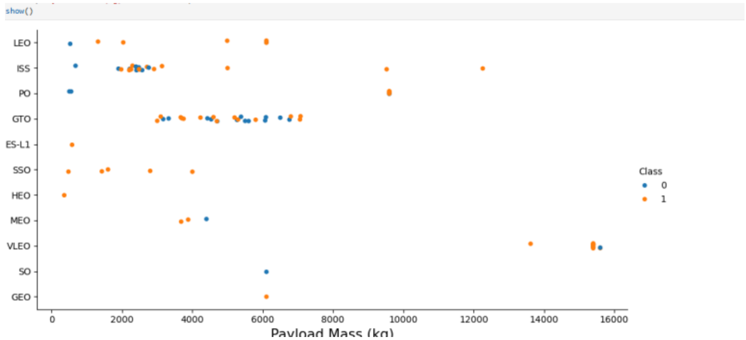
Future work

Add weather, launch window, mission complexity and newer booster reuse history to improve predictive power.

Appendix: Additional EDA Visuals







Additional visuals were used during analysis to compare flight number, payload mass, launch site and orbit type against landing success. These visuals support the final interpretation that launch success is multi-factorial and has improved with experience.

Appendix: Project Deliverables Checklist

- ✓ **GitHub repository link**
- ✓ **Completed notebooks and Python files**
- ✓ **PDF presentation**
- ✓ **EDA, SQL, Folium, Dash and ML results**
- ✓ **Creative/innovative insights**

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